1 VISUALIZING DATA (PART 1)

1.1 ONE-WAY TABLES

1.1.1 INDIVIDUALS AND VARIABLES

The simplest kind of information we'll work with in this course is a set of **individuals** with one or more properties, called **variables**. The individuals are the items in the data set and can be cases, things, people, etc.

In this table for ***heights of people***, the individuals are the people, and their heights are a variable because height is a property of each individual.

Person	Height
Wendy	5'6"
Michael	5'9"
Rachael	5'3"
Allen	5'11"

Together, individuals and variables are called data. When we organize **data** into a table like this one, we call it a **data table**. Here's an example of a data table about ice cream:

Flavor	Scoops Sold	Contains chocolate?	Smooth or chunky?
Vanilla	300	No	Smooth
Chocolate	450	Yes	Smooth
Cookies & Cream	275	Yes	Chunky
Mint Chocolate Chip	315	Yes	Chunky
Fudge Brownie	375	Yes	Chunky
Rocky Road	250	Yes	Chunky

The individuals are the flavors: Vanilla, Chocolate, Cookies & Cream, etc., and the variables are their properties: Scoops sold, Contains chocolate, and Smooth or chunky.

Variables can be **categorical** or **quantitative**. In Error! Reference source not found., there's one quantitative variable: the height. In **Error! Reference source not found.**, there's one quantitative variable (scoops), and two categorical variables (contains chocolate and smooth or chunky).

- **Categorical variables** are non-numerical variables. Categorical variables are also called ***qualitative*** variables. Their values aren't represented with numbers. Whether or not the ice cream contains chocolate, and whether the ice cream is smooth or chunky are categorical variables. This is because the values there are words, not numbers.

- **Quantitative variables** are numerical variables. Their values are numbers. The height of the people and the number of scoops of ice cream sold are quantitative variables, because the values there are numbers.

We can divide quantitative variables into **discrete variables** and **continuous variables**.

- **Discrete variables** are those that we can obtain by counting. Therefore, they can take on only certain numerical values. For example, the number of scoops of ice cream sold is a discrete quantitative variable, because we can't really sell 8.3 or 5.23 scoops of ice cream.

- On the other hand, **continuous variables** may include data as decimals, fractions, or irrational numbers. For example, the height of the people in the car is a continuous variable.

1.1.2 LEVELS OF MEASUREMENT

When we work with data, we need to understand the level of measurement, because not every statistical test can be used with every type of data set. There are four levels of data measurement: **nominal**, **ordinal**, **interval**, and **ratio**.

- Things like favorite food, colors, names, and "yes" or "no" responses have a **nominal** scale of measurement. Only categorical data can be measured with a nominal scale.

- Categorial data can also be **ordinal**. This type of data can be ordered. The top three national parks in Philippines is an example of ordinal data. Or, for example, when we ask a group of people about how they liked their trip, we may get responses like "awesome", "good", "satisfactory", or "terrible", or in numerical form (4,3,2,1), which follow an order from best to worst.

- Data measured using an **interval** scale can be ordered like ordinal data. But interval data also gives us a known interval between measurements. For example, temperature is measured using an interval scale, because we can understand the exact interval of difference between 50 and 60 degrees.

- Data measured using a **ratio** scale is just like interval scale data, except that ratio scale data has a starting point, or absolute zero. Whereas interval data like temperature can have negative and positive values, things like time, height, and weight are examples of ratio scale data because those measures can never be negative.

When we have a single individual and a categorical or quantitative variable assigned to it, we can construct a one-way table. These kinds of tables are called ***one-way*** because they usually represent data for one individual only.

1.1.3 CONSTRUCTING ONE-WAY TABLES

When we construct a table, we want to think about whether we have more individuals or more variables. In the two tables we looked at, we had the individuals listed down the left-hand side, and the variables listed across the top.

But if we have lots of variables but only a few individuals, it can be helpful to list the individuals across the top and the variables down the left side.

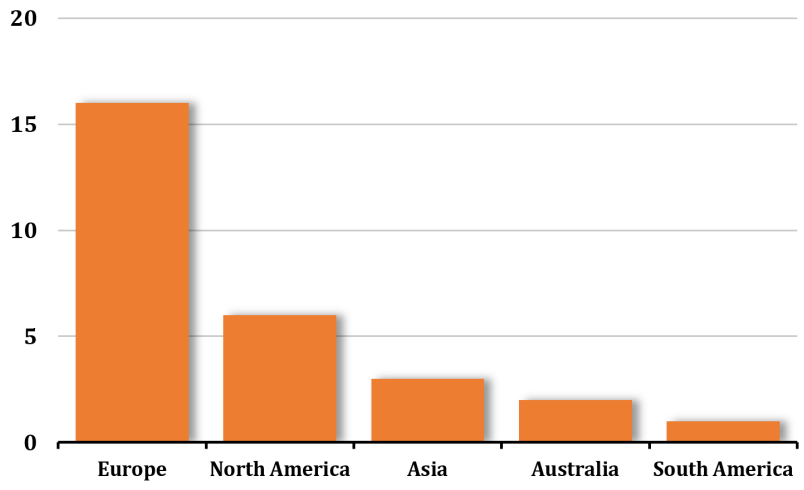
For example, if we're comparing lots of information about two houses for sale in the same neighborhood, we might make a table like this:

	317 Molave St	819 Camagong St
Price	₱16,800,000	₱19,500,000
Square meter	339	354
Price per square meter	₱49,560	₱55,085
Bedroom	4	5
Bathrooms	3	3
Storeys	3	2
Basement	Finished	Unfinished
Garage Spaces	3	2
Lot area (sqm)	300	288
Grass Backyard	Yes	Yes
Year built	2005	2018
Property Tax	₱18,500	₱22,000
Payment	₱78,000	₱72,000

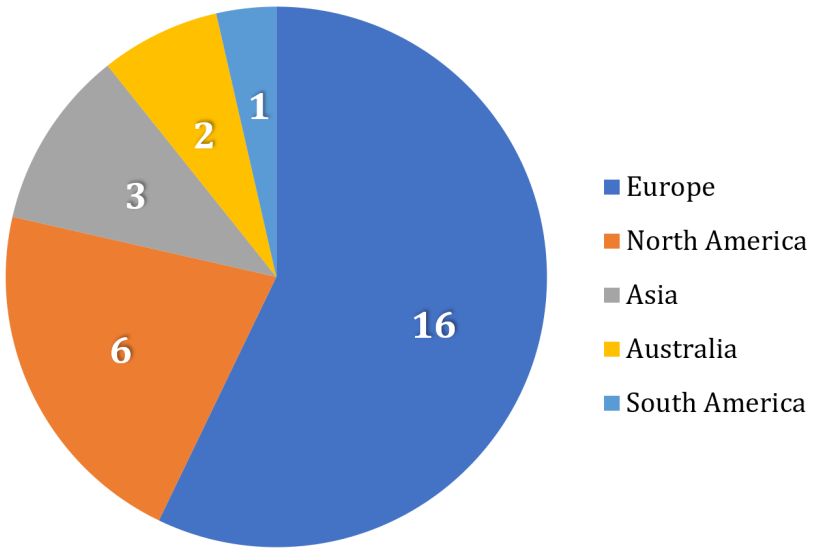
The individuals are the addresses listed across the top of the table, **317 Molave St** and **819 Camagong St**. All the variables are listed down the left side of the table. Since there are so many more variables than individuals, listing the variables vertically makes the table fit better on paper than if we had tried to list all the variables horizontally.

1.2 BAR GRAPHS AND PIE CHARTS

Bar graphs and pie charts are two of the simplest ways to summarize and represent data. In general, a **bar graph**, also called a **bar chart**, usually looks something like this:



And a **pie chart** usually looks something like this:



In this lesson, we'll start with data tables and try to represent the data given in the tables in bar graphs and pie charts.

1.2.1 BUILDING DATA TABLES

Let's say we're given historical information about the host cities for the summer Olympic games, and we want to summarize this information into a simple data table.

Here is a list of host cities for the summer games, not including host cities for canceled games, from 1896 through 2016.

Games	Year	City, Country	Continent
I	1896	Athens, Greece	Europe
II	1900	Paris, France	Europe
III	1904	St. Louis, United States	North America
IV	1908	London, United Kingdom	Europe
V	1912	Stockholm, Sweden	Europe
VII	1920	Antwerp, Belgium	Europe
VIII	1924	Paris, France	Europe
IX	1928	Amsterdam, Netherlands	Europe
X	1932	Los Angeles, United States	North America
XI	1936	Berlin, Germany	Europe
XIV	1948	London, United Kingdom	Europe
XV	1952	Helsinki, Finland	Europe
XVI	1956	Melbourne, Australia	Australia
XVII	1960	Rome, Italy	Europe
XVIII	1964	Tokyo, Japan	Asia
XIX	1968	Mexico City, Mexico	North America
XX	1972	Munich, West Germany	Europe
XXI	1976	Montreal, Canada	North America
XXII	1980	Moscow, Soviet Union	Europe
XXIII	1984	Los Angeles, United States	North America
XXIV	1988	Seoul, South Korea	Asia
XXV	1992	Barcelona, Spain	Europe
XXVI	1996	Atlanta, United States	North America
XXVII	2000	Sydney, Australia	Australia
XXVIII	2004	Athens, Greece	Europe
XXIX	2008	Beijing, China	Asia
XXX	2012	London, United Kingdom	Europe
XXXI	2016	Rio de Janeiro, Brazil	South America

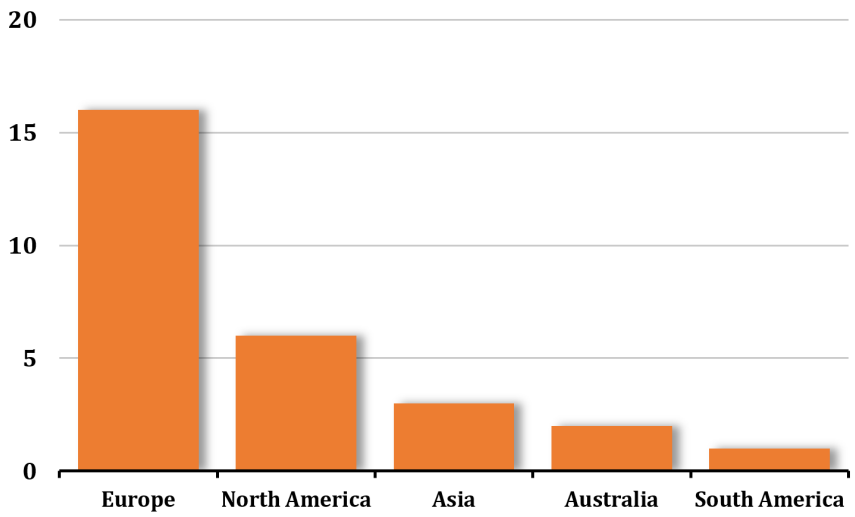
If we wanted to make a data table showing the number of times each continent has hosted to summer games, we could count this number for each continent from the data table, and create a summary table for count by continent:

Continent	Count
Europe	14
North America	6
Asia	3

The summary table is often called a **frequency table**, which shows the frequency or count of each individual. In the list of host cities of summer games, the individual, Europe, appeared 16 times, which is why the count, or frequency is **16**.

1.2.2 BUILDING BAR GRAPHS

If we wanted to express the count of summer games by continent in a bar graph, it might look like this:

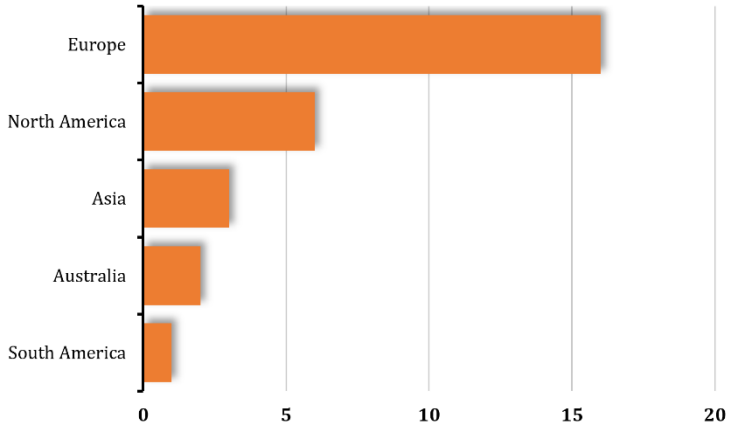


Notice that we have a list of the continents across the bottom of the bar graph, with the count of the number of times they've hosted the summer games up the left side. The continents are the **individuals**, and the count is a **quantitative variable**, because the count is a numeric property of each of the individuals.

The bar graph is a nice way to represent this data, because we can quickly get a visual picture of which continents have hosted the summer games most often.

Now we can quickly see that Europe has hosted more summer games by far than any other continent, North America has hosted the second-most number, and South America has hosted the summer games the fewest number of times. With this particular data set, since we know there are 7 continents, we could infer from the graph that Africa and Antarctica have never hosted the summer games.

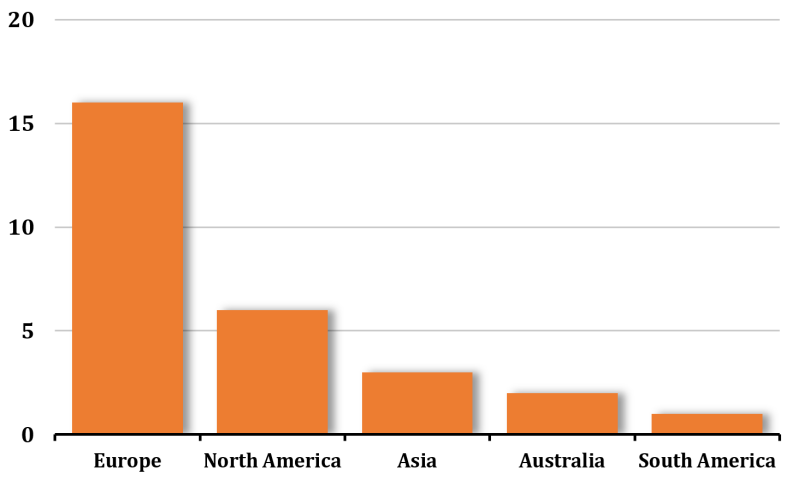
Bar graphs can also be built horizontally.



When we build a vertical bar graph, it's common to sort the data largest to smallest, so that the tallest bars appear on the left, in descending order down to the smallest bars on the right. When we build a horizontal bar graph, it's common to put the largest bars at the top and the smallest bars at the bottom.

1.2.3 READING BAR GRAPHS

If we only have the bar graph, and no data table to work with, we may only be able to get approximate values from the bar graph. Using this bar graph again,



we see that the vertical axis isn't marked off at every increment, only at every increment of **5**. So based on how far up the bar extends for Europe, for example, we only know with absolute certainty that Europe has hosted between **15** and **20** times. We could probably guess that they've hosted about **16** or **17** summer games, but we might not feel absolutely sure.

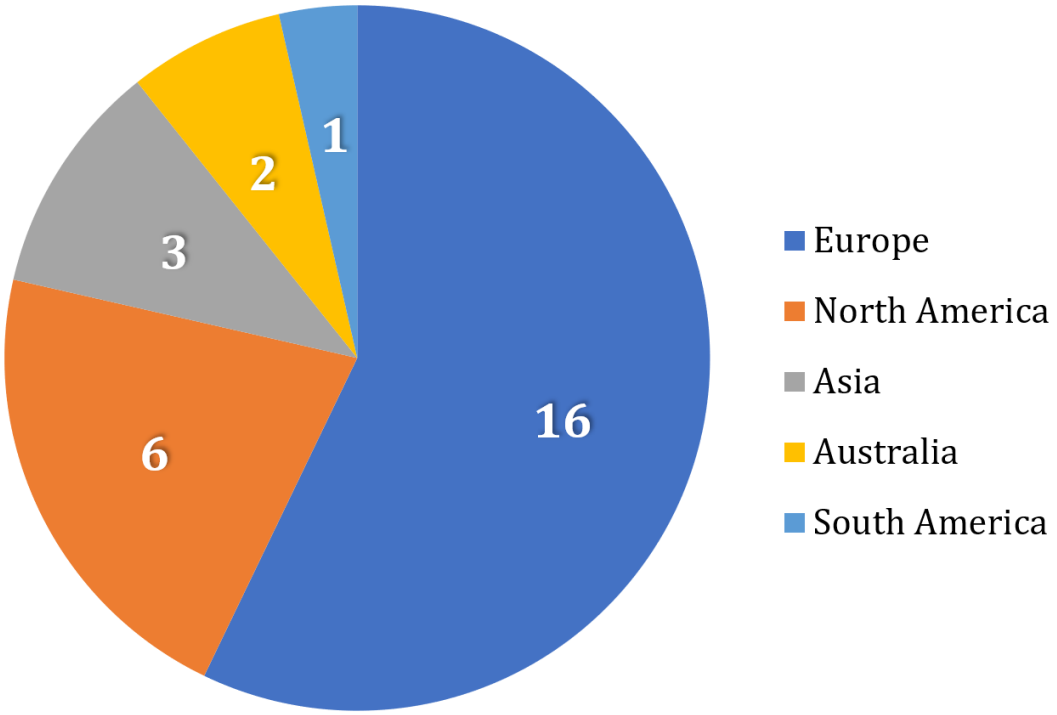
At quick glance we would know that Australia has probably hosted only **2** or **3** times

It's worth making the point that bar graphs aren't always great at expressing exact values, but they're excellent at giving us a quick visual picture of data

1.2.4 BUILDING PIE CHARTS

This same data can be displayed in a pie chart. It's helpful in a pie chart, though not necessary, to make each individual a different color so that it's easy to see the distinction between sections.

Like the bar graphs, it's customary to put the largest slices next to each other, in order all the way down to the smallest sections.



Example

Create a bar graph and pie chart that shows the number of times each continent has hosted the winter Olympic games. Use the data table to first create a summary table, then build the bar graph and pie chart.

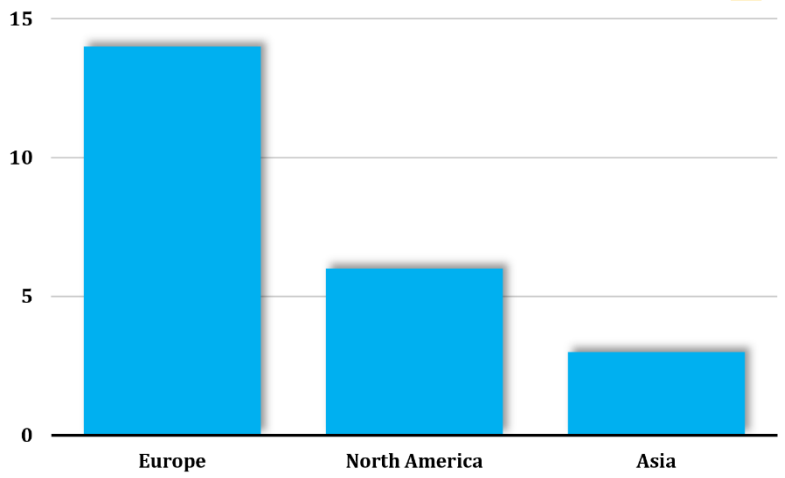
Host cities for the winter Olympic games, not including host cities for canceled games, from 1924 through 2018

Games	Year	City, Country	Continent
I	1924	Chamonix, France	Europe
II	1928	St. Moritz, Switzerland	Europe
III	1932	Lake Placid, United States	North America
IV	1936	Garmisch-Partenkirchen, Germany	Europe
V	1948	St. Moritz, Switzerland	Europe
VI	1952	Oslo, Norway	Europe
VII	1956	Cortina d'Ampezzo, Italy	Europe
VIII	1960	Squaw Valley, United States	North America
IX	1964	Innsbruck, Austria	Europe
X	1968	Grenoble, France	Europe
XI	1972	Sapporo, Japan	Asia
XII	1976	Innsbruck, Austria	Europe
XIII	1980	Lake Placid, United States	North America
XIV	1984	Sarajevo, Yugoslavia	Europe
XV	1988	Calgary, Canada	North America
XVI	1992	Albertville, France	Europe
XVII	1994	Lillehammer, Norway	Europe
XVIII	1998	Nagano, Japan	Asia
XIX	2002	Salt Lake City, United States	North America
XX	2006	Turin, Italy	Europe
XXI	2010	Vancouver, Canada	North America
XXII	2014	Sochi, Russia	Europe
XXIII	2018	Pyeongchang, South Korea	Asia

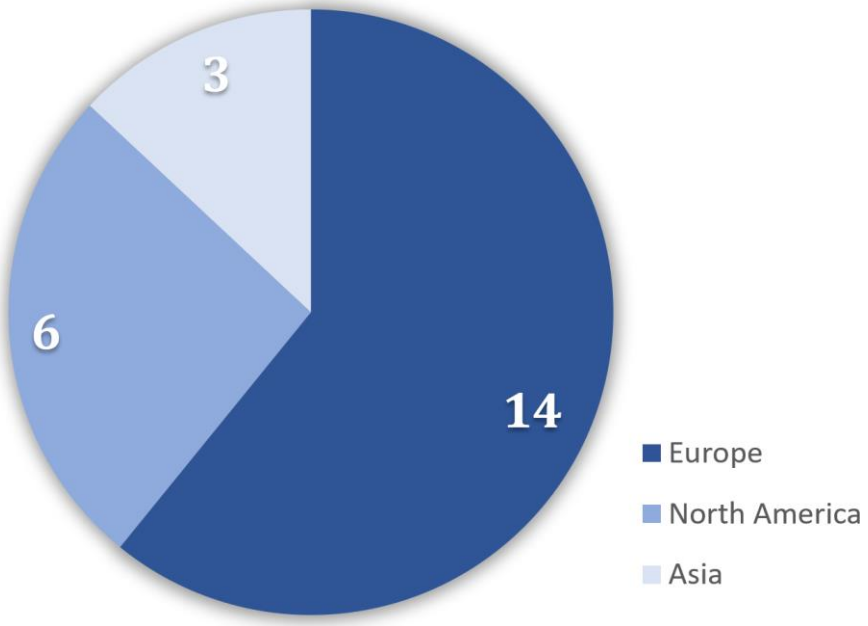
If we try to make a frequency table first of this information, we get

Continent	Count
Europe	14
North America	6
Asia	3

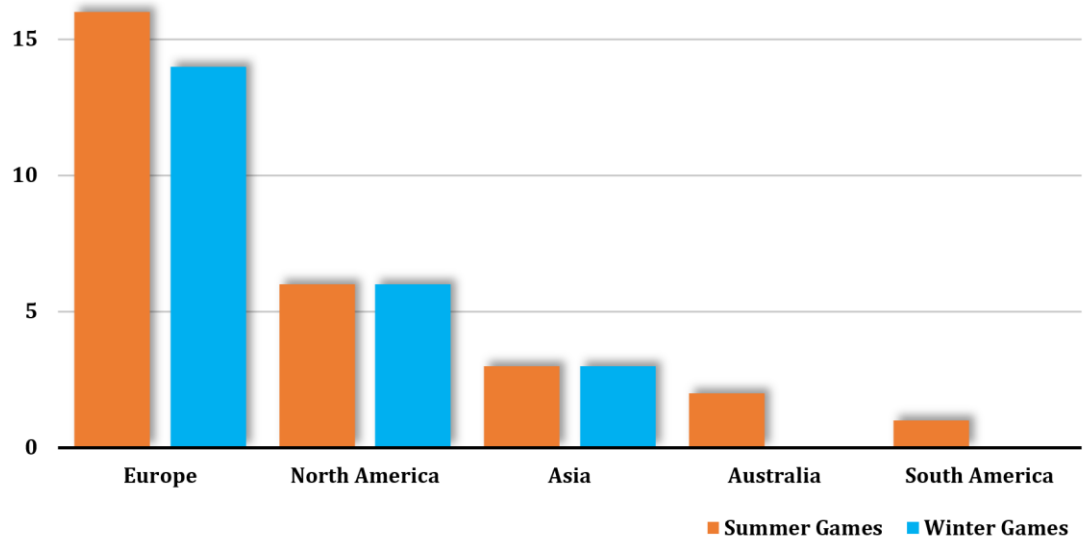
Then we can make a bar graph,



or a pie chart.



Bar graphs are also great for showing multiple variables for the same individuals, side by side. Now that we've created bar graphs for host continents for both the summer and winter Olympic games, let's bring them together:



This kind of side-by-side bar graph allows us to quickly see what we already knew from the previous bar graphs, like the fact that Europe has hosted more summer games and more winter games than any other continent.

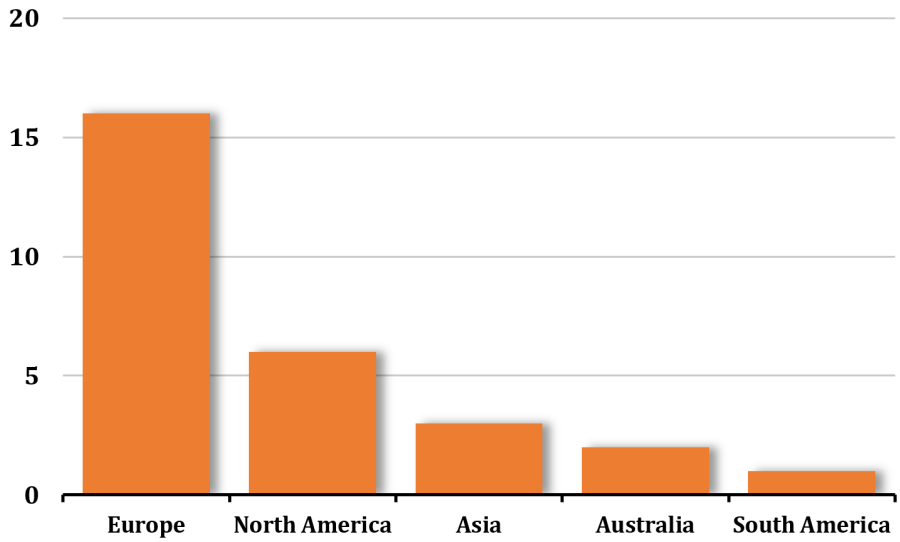
But we get more information from this, too, like the fact that Europe has hosted more summer games than it has winter games, or that North America has hosted an equal number of summer and winter games.

1.3 LINE GRAPHS AND OGIVES

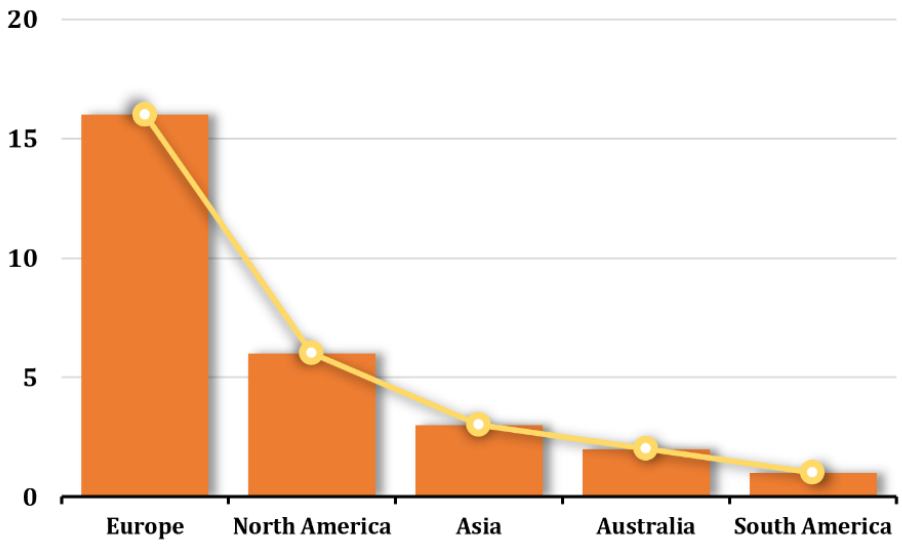
1.3.1 LINE GRAPHS

Line graphs are really similar to bar graphs. In fact, to turn a bar graph into a line graph, all we have to do is connect the middle of the top of each bar to the middle of the top of the bar beside it with a straight line, and we'll form the line graph.

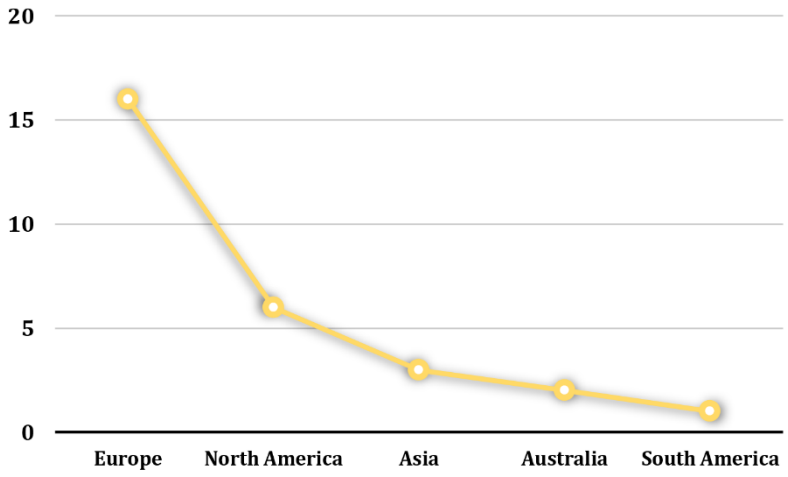
For example, we could take a bar graph for the summer Olympics from the previous section.



Then we can use straight lines to connect the top of each bar.



Then we can remove the bars and we have a line graph that represents the same data.

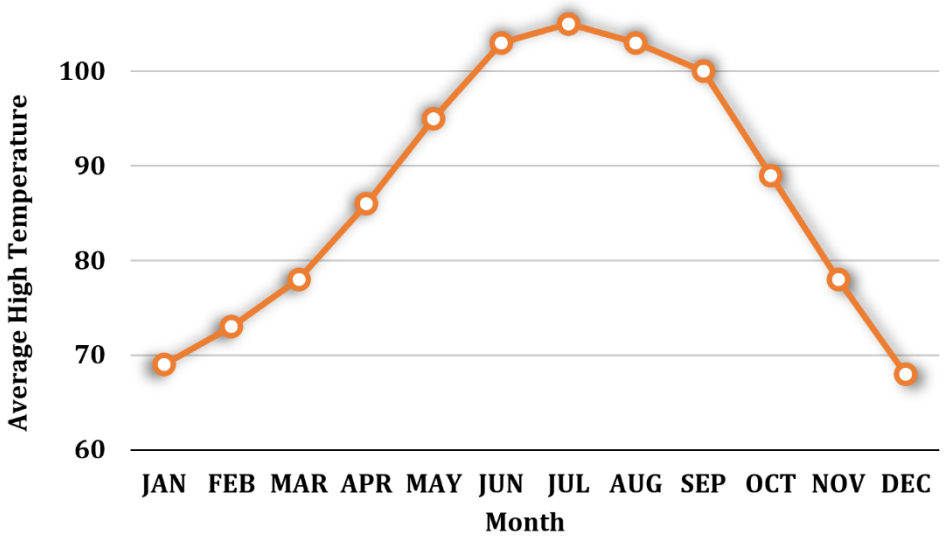


While this example is helpful for illustrating how we can transition from a bar graph to a line graph, and how the two graph types are related, in actuality, a line graph is a bad choice for this data. Line graphs are used to show changes over time or a connection between individuals, but this data is comparing things between different groups. That means the bar graph is a better choice than a line graph for this data in particular.

Let's look at an example with data that's much better suited for a line graph, like the monthly average high temperature in Scottsdale, Arizona over the course of a year. The data is

Month	Average high
JAN	69
FEB	73
MAR	78
APR	86
MAY	95
JUN	103
JUL	105
AUG	103
SEP	100
OCT	89
NOV	78
DEC	68

The historical average high temperature in degrees Fahrenheit is given for each month. If we plot these as a line graph, we get



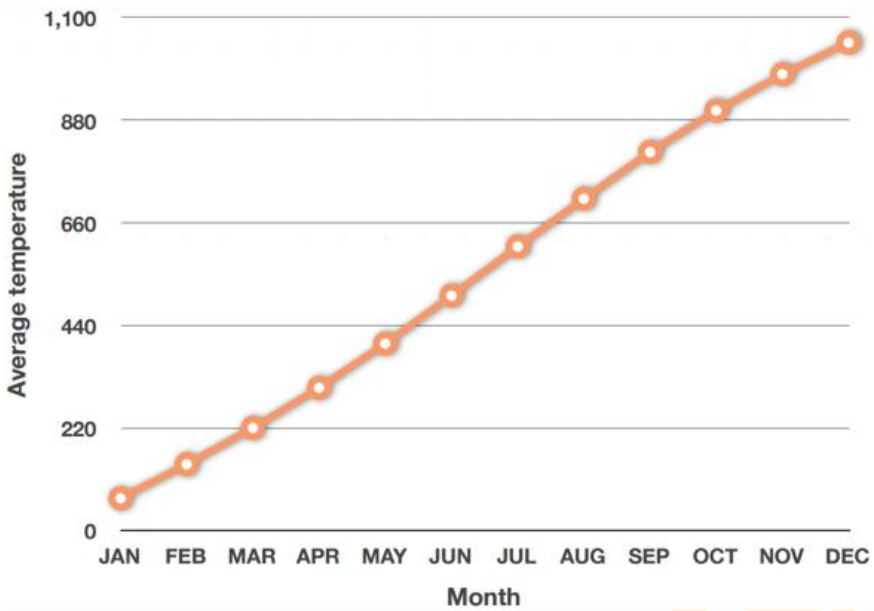
Notice that the slope of each line segment gives us an idea of how much change occurred between two data points. For example, there's a much larger change in average high temperature between September and October than there is between August and September, because the slope of the line connecting September to October is much steeper, indicating a larger decrease in temperature, than the slope of the line connecting August to September, which is much flatter, and therefore indicates a smaller decrease in temperature.

A line graph is a fantastic way to display this kind of data, mostly because it shows clearly how a data point changes over time. We inherently understand that average high temperature fluctuates over the course of a year, with (in the northern hemisphere) the highest temperatures usually occurring in the summer months, and the lowest temperatures usually occurring the winter months.

1.3.2 OGIVES

An **ogive** is a special kind of line graph. This kind of graph looks just like a line graph, but think of an ogive as an "accumulated" line graph. Just like other types of graphs, an ogive does well at representing some kinds of data, and less well at representing others.

For example, it doesn't make sense to show accumulated temperature over time. Taking the average monthly temperature in Scottsdale, Arizona, if we use an ogive, we would show the temperature in January as 69, the temperature in February as $69 + 73$, the temperature in March as $69 + 73 + 78$ etc. The ogive would look like this:

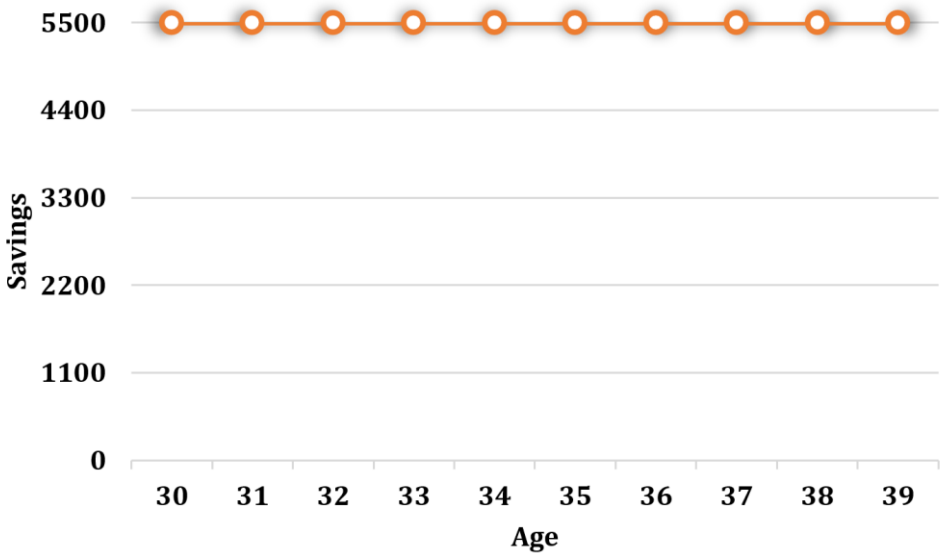


But it doesn't make sense to have a temperature of almost **1,100** degrees in December. Temperature is not a very good data point to represent with an ogive.

But an ogive would be perfect for showing something like accumulated savings over time. For example, let's say we invest **₱5,500** each year into a savings account, from age **30** until age **39**.

Age	Investment
30	₱5,500
31	₱5,500
32	₱5,500
33	₱5,500
34	₱5,500
35	₱5,500
36	₱5,500
37	₱5,500
38	₱5,500
39	₱5,500

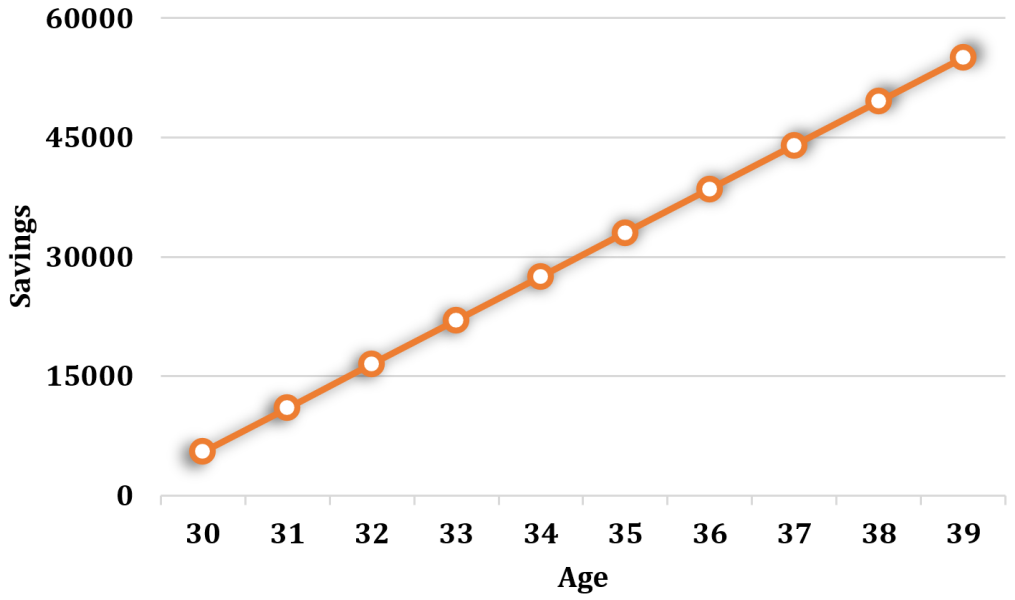
Graphing the invested amount each year in a line graph gives



While this is valuable information, we might be even more interested in the accumulated amount over time. If we find the accumulated total after each year, the data is

Age	Investment	Total
30	₱5,500	₱5,500
31	₱5,500	₱11,000
32	₱5,500	₱16,500
33	₱5,500	₱22,000
34	₱5,500	₱27,500
35	₱5,500	₱33,000
36	₱5,500	₱38,500
37	₱5,500	₱44,00
38	₱5,500	₱49,500
39	₱5,500	₱55,000

Graphing the accumulated total each year could be much more valuable, because it gives us visibility into how our savings will grow over time.



From this ogive, we can quickly see that we'll have something between **₱50,000** and **₱60,000** once we've made our investment at age **39**. Or we can see that we'll cross the **₱30,000** mark once we make our investment at age **35**.